Hook: Classifying the Metropolitan Museum of Art's Paintings by Historical Period

A DS 4002 Case Study by Carly Farrey

Bridge over a Pond of Water Lilies - Claude Monet

Museums are home to thousands of works of art created throughout history. These works are often classified by the period of time in which they were painted, including Renaissance and Modern. To classify these time periods, one must know about art history including specifics about techniques used, color palettes, and stylistic trends. Classifications like these are now possible using machine learning. Computers can analyze visual patterns found in images and learn these distinctions, without requiring extensive knowledge of art.

You are a data scientist working with digital paintings from the Metropolitan Museum of Art's website. The Met has a large dataset that is full of paintings and metadata that describes the time period and other basic facts about each painting. The Met wants to know if a machine learning model can accurately classify the paintings by their time periods based only on the visual features present in the image.

Your task is to build a classification model that predicts the historical period a painting is from. You will use images from the Met dataset, preprocess the images, organize the images by period, and then train a convolutional neural network (CNN) to learn the patterns in the paintings. Then, you will assess the models performance using accuracy and a confusion matrix. This will determine if a machine learning model can reliably classify paintings by historical periods based only on image features.

The Metropolitan Museum of Art. (1899). *Bridge over a Pond of Water Lilies*. The Metropolitan Museum of Art. https://www.metmuseum.org/art/collection/search/437127